

CSE 220 · Signals and Linear Systems Sessional

FourierLab

Exploring Signals in the Frequency Domain

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Audio Lab

STFT · Time-Frequency

Image Lab

2D DFT · k-Space MRI

Music Lab

Source Sep. · DSP Mixing

Waveform & Spectral Analysis

Core Transform

The Short-Time Fourier Transform slides a window $w(n)$ over the signal, computing a DFT per frame — yielding magnitude and phase at every (time, frequency) bin.

$$X(m,k) = \sum x(n+mH) \cdot w(n) \cdot e^{-j2\pi kn/N}$$

m = frame H = hop length N = FFT size

Magnitude vs. Phase

Reconstruct using magnitude-only (zero phase) and phase-only (unit magnitude). Magnitude carries the perceptual content — phase alone sounds like noise.

$$X(m,k) = |X(m,k)| \cdot e^{j \angle X(m,k)}$$

STFT

Windowing

ISTFT

Spectrogram



Masking · Retention · Filter & EQ

Lasso Masking

Draw freeform polygons directly on the spectrogram to Remove or Isolate time-frequency regions.

$$\text{mask}(m,k) = 0 \quad \text{if } (t,f) \in \text{region}$$

SNR · MSE · Spectral Convergence

Point-in-polygon ray-casting per (time, freq) bin

Progressive Retention

Keep only the top-N magnitude bins or lowest frequencies. Sweep retention fraction 2–100% and plot SNR vs. fraction retained.

$$D_{\text{ret}} = \text{top-}k(|X|) \quad \text{or} \quad \text{low-freq}(X)$$

Strategies: Low Frequency · Top Magnitude

Concept: energy compaction — most signal energy lives in a small subset of coefficients

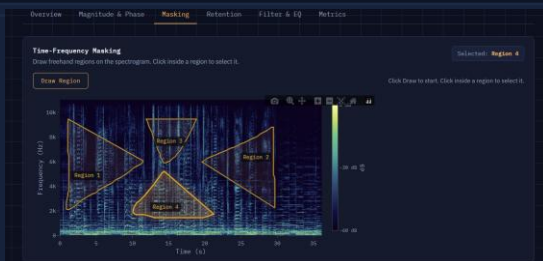
Filter & EQ Chain

Stack Lowpass · Highpass · Bandpass · Notch · Custom Gain blocks, each with a freq range and optional time range.

$$Y(m,k) = X(m,k) \cdot G(k) \cdot 1[t \in \text{range}]$$

Cosine-tapered edges to prevent Gibbs ringing

Real-time EQ curve preview (gain dB vs frequency)



MRI k-Space Reconstruction

2D DFT → k-Space

MRI acquires raw data in the frequency domain (k-space). Low spatial frequencies near center encode contrast; high frequencies at periphery encode edges.

$$K(u,v) = \sum_x \sum_y f(x,y) \cdot e^{-j2\pi(ux/M + vy/N)}$$

Hero Feature – Lasso Masking on k-Space

Draw a freeform polygon directly on the k-space image. Ray-casting in normalized coordinates $(k_x, k_y) \in [-0.5, 0.5]^2$. Isolate or remove the region, then reconstruct via IFFT2.

Progressive Retention

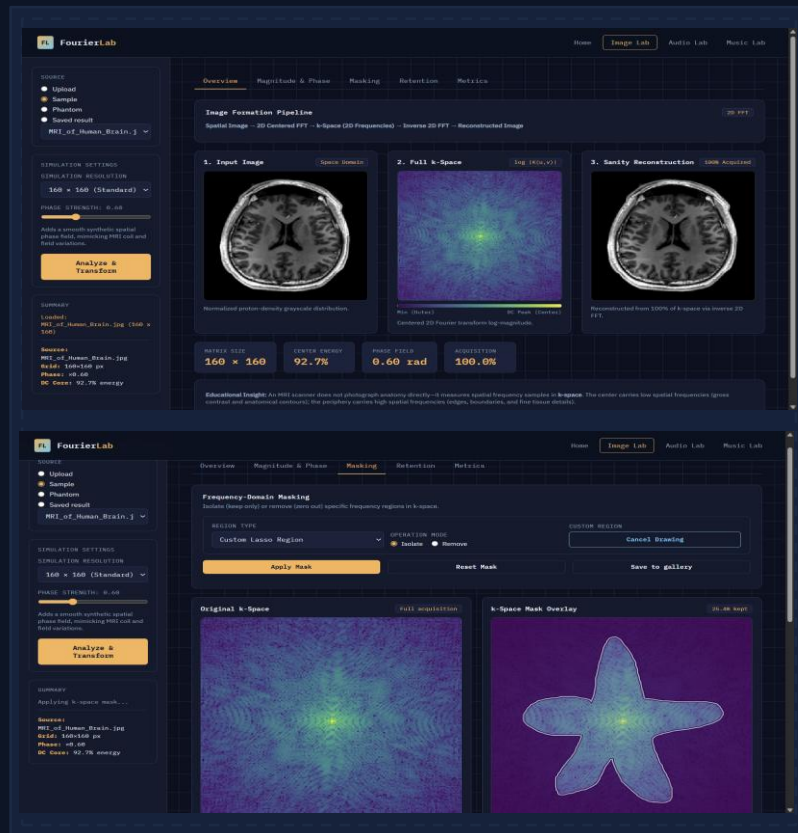
Retain center-radius fraction or top-magnitude k-space bins. Sweep → PSNR & SSIM vs. retention %.

MSE

PSNR

SSIM

Error Heatmap



Source Separation & Audio Mixing

Vocal / Instrumental Separation

Uses Demucs — Meta's hybrid transformer model (waveform + spectrogram domain). Separates audio into vocal and instrumental stems in two-stem mode.

Neural network trained on paired data — goal matches freq-domain masking, but learned rather than hand-crafted.

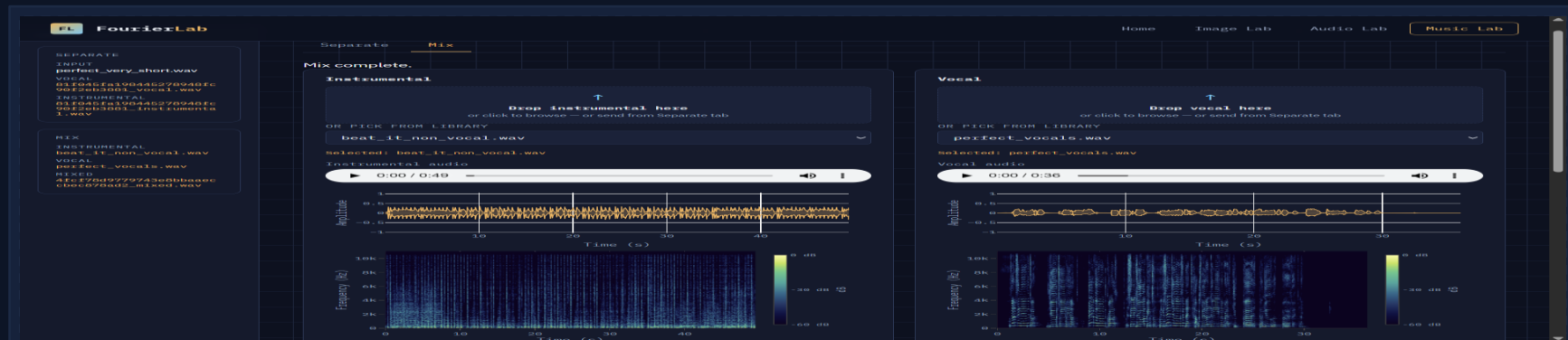
ML Model

DSP Mixing

Auto-align tracks via cross-correlation, apply independent gain per stem, then mix by superposition:

$$y(t) = g_{\text{inst}} \cdot x_{\text{inst}}(t) + g_{\text{voc}} \cdot x_{\text{voc}}(t - \tau)$$

$$\tau = \operatorname{argmax}(x_{\text{inst}} * x_{\text{voc}}) \quad \cdot \quad g = 10^{(\text{GdB}/20)}$$



Signals & Linear Systems Concepts Applied

FEATURE	MODULE	CONCEPT
Waveform & STFT Spectrogram	Audio Lab	Short-Time Fourier Transform, Windowing
Magnitude vs. Phase	Audio Lab	Polar form of complex spectrum
Lasso Masking	Audio Lab	Time-frequency masking, ISTFT reconstruction
Progressive Retention	Audio Lab	Energy compaction, signal approximation
Filter & EQ Chain	Audio Lab	Freq-selective filtering, Convolution theorem
2D DFT & k-Space Analysis	Image Lab	2D Discrete Fourier Transform, spatial frequency
Lasso Masking on k-Space	Image Lab	2D freq masking, Inverse 2D DFT
MRI Retention Sweep	Image Lab	Circular low-pass filtering in k-space
Demucs Separation	Music Lab	ML-based source separation (beyond DSP)
Audio Mixing	Music Lab	Superposition, cross-correlation, gain

"Transform · Mask · Reconstruct · Measure – one framework, every domain."